

EXPERIMENT 7

Design and configure the hub network using cisco packet tracer

AIM:

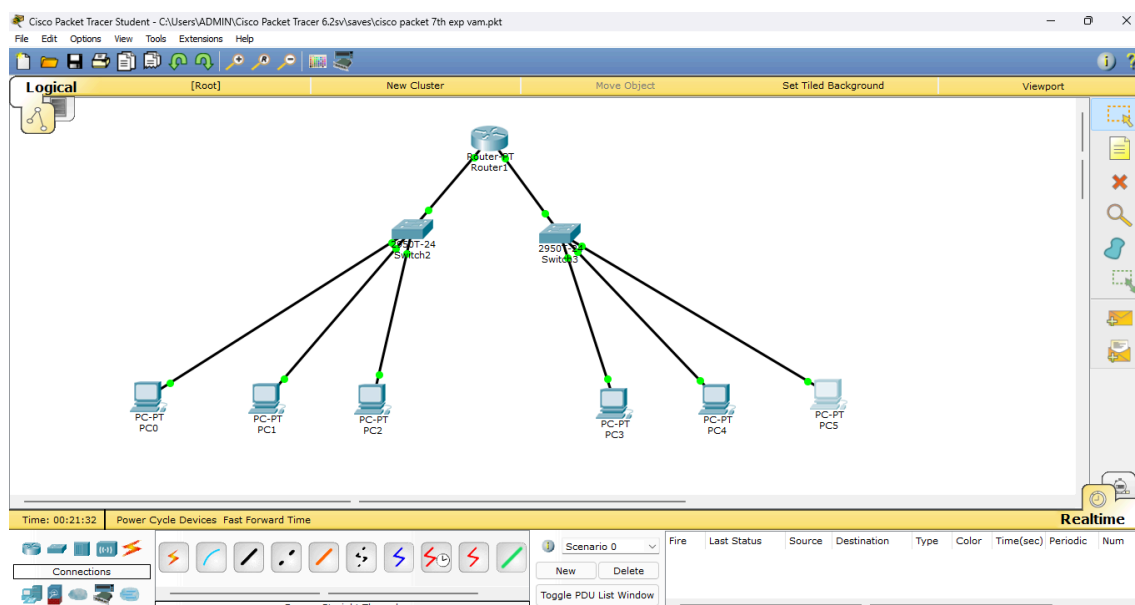
To divide the IP block 192.168.10.0/25 into two subnets that can support at least 3 hosts each.

To configure the subnets and verify connectivity between computers using PDU (ping) packets.

PROCEDURE :

1. Determine subnetting: Split 192.168.10.0/25 into two /26 subnets → Subnet 1: 192.168.10.0/26, Subnet 2: 192.168.10.64/26.
2. Identify host ranges:
3. Subnet 1: 192.168.10.1 – 192.168.10.62 (Broadcast: 192.168.10.63)
4. Subnet 2: 192.168.10.65 – 192.168.10.126 (Broadcast: 192.168.10.127)
5. Assign IP addresses to at least 3 computers in each subnet and configure subnet mask (255.255.255.192).
6. Connect systems using a switch/router (if inter-subnet communication is needed, configure router interfaces with gateway IPs).
7. Test connectivity by sending PDU (ping) packets within and between subnets to verify communication.

BLOCK DIAGRAM :



OUTPUT :

The screenshot shows a web-based configuration interface for a router named "Router1". The interface has three tabs: "Physical", "Config", and "CLI". The "Config" tab is active, showing a tree view on the left with categories: GLOBAL (Settings, Algorithm Settings), ROUTING (Static, RIP), and INTERFACE (FastEthernet0/0, FastEthernet1/0, Serial2/0, Serial3/0, FastEthernet4/0, FastEthernet5/0). The "FastEthernet0/0" interface is selected, and its configuration is displayed on the right. The configuration includes: Port Status (On), Bandwidth (100 Mbps), Duplex (Full Duplex), MAC Address (0005.5E2B.805B), IP Configuration (IP Address: 192.168.10.4, Subnet Mask: 255.255.255.128), and Tx Ring Limit (10). Below the configuration, there is a section titled "Equivalent IOS Commands" containing a list of commands to replicate the configuration in IOS.

Router1

Physical Config CLI

FastEthernet0/0

Port Status ☒ On

Bandwidth ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 0005.5E2B.805B

IP Configuration

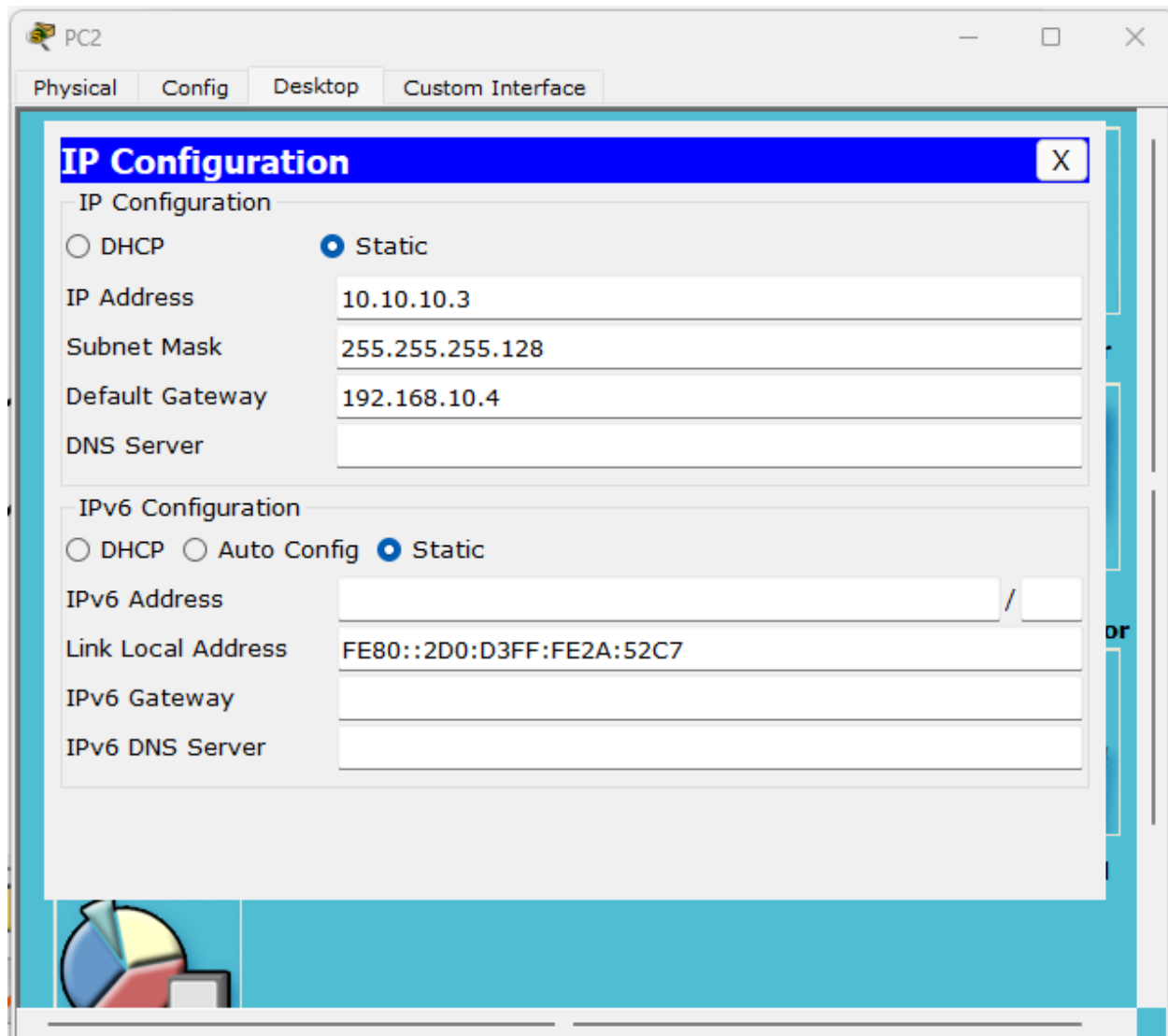
IP Address 192.168.10.4

Subnet Mask 255.255.255.128

Tx Ring Limit 10

Equivalent IOS Commands

```
Router(config)#exit
Router(config)#interface FastEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet1/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/0
Router(config-if)#
```



RESULT :

The network was successfully divided into two subnets with proper IP allocation. Connectivity between computers within and across subnets was verified using PDU packets.